

September 24, 2021

Cindy Tomaszewski
City of Cadillac
1121 Plett Rd
Cadillac, MI 49601

RE: Project: Fall Biosolids
Pace Project No.: 50296623

Dear Cindy Tomaszewski:

Enclosed are the analytical results for sample(s) received by the laboratory on September 03, 2021. The results relate only to the samples included in this report. Results reported herein conform to the applicable TNI/NELAC Standards and the laboratory's Quality Manual, where applicable, unless otherwise noted in the body of the report.

The test results provided in this final report were generated by each of the following laboratories within the Pace Network:

- Pace Analytical Services - Indianapolis

Revised Report: This report replaces the original dated, 092321. Revised to change mercury results due to lab reporting error./092421msb

If you have any questions concerning this report, please feel free to contact me.

Sincerely,



Melanie Booms
melanie.booms@pacelabs.com
(616)975-4500
Project Manager

Enclosures



REPORT OF LABORATORY ANALYSIS

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CERTIFICATIONS

Project: Fall Biosolids

Pace Project No.: 50296623

Pace Analytical Services Indianapolis

7726 Moller Road, Indianapolis, IN 46268

Illinois Accreditation #: 200074

Indiana Drinking Water Laboratory #: C-49-06

Kansas/TNI Certification #: E-10177

Kentucky UST Agency Interest #: 80226

Kentucky WW Laboratory ID #: 98019

Michigan Drinking Water Laboratory #9050

Ohio VAP Certified Laboratory #: CL0065

Oklahoma Laboratory #: 9204

Texas Certification #: T104704355

Wisconsin Laboratory #: 999788130

USDA Soil Permit #: P330-19-00257

REPORT OF LABORATORY ANALYSIS

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SAMPLE SUMMARY

Project: Fall Biosolids

Pace Project No.: 50296623

Lab ID	Sample ID	Matrix	Date Collected	Date Received
50296623001	Digestor #4	Solid	09/02/21 14:00	09/03/21 09:15

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SAMPLE ANALYTE COUNT

Project: Fall Biosolids

Pace Project No.: 50296623

Lab ID	Sample ID	Method	Analysts	Analytes Reported	Laboratory
50296623001	Digestor #4	EPA 6010	JPk	6	PASI-I
		EPA 6020	CAW	10	PASI-I
		EPA 7471	FRW	1	PASI-I
		SM 2540G	WZE	1	PASI-I
		SM 2540G	WZE	1	PASI-I
		SM 2540G	WZE	1	PASI-I
		EPA 9045	HCF	1	PASI-I
		EPA 351.2	MMS	1	PASI-I
		EPA 353.2	GWA	2	PASI-I
		SM 4500-NH3 G	MMS	1	PASI-I
		SM 4500-CI-E	GWA	1	PASI-I

PASI-I = Pace Analytical Services - Indianapolis

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ANALYTICAL RESULTS

Project: Fall Biosolids
Pace Project No.: 50296623

Sample: Digestor #4 **Lab ID: 50296623001** Collected: 09/02/21 14:00 Received: 09/03/21 09:15 Matrix: Solid

Results reported on a "dry weight" basis and are adjusted for percent moisture, sample size and any dilutions.

Parameters	Results	Units	Report Limit	DF	Prepared	Analyzed	CAS No.	Qual
6010 MET ICP								
Analytical Method: EPA 6010 Preparation Method: EPA 3050								
Pace Analytical Services - Indianapolis								
Aluminum	3920	mg/kg	3430	1	09/21/21 18:45	09/22/21 00:18	7429-90-5	
Calcium	17900	mg/kg	3430	1	09/21/21 18:45	09/22/21 00:18	7440-70-2	
Magnesium	3480	mg/kg	3430	1	09/21/21 18:45	09/22/21 00:18	7439-95-4	
Phosphorus	24500	mg/kg	3430	1	09/21/21 18:45	09/22/21 00:18	7723-14-0	N2
Potassium	<3430	mg/kg	3430	1	09/21/21 18:45	09/22/21 00:18	7440-09-7	
Sodium	4110	mg/kg	3430	1	09/21/21 18:45	09/22/21 00:18	7440-23-5	
6020 MET ICPMS								
Analytical Method: EPA 6020 Preparation Method: EPA 3050B								
Pace Analytical Services - Indianapolis								
Arsenic	21.4	mg/kg	1.0	1	09/07/21 19:48	09/08/21 13:35	7440-38-2	
Cadmium	1.7	mg/kg	0.51	1	09/07/21 19:48	09/08/21 13:35	7440-43-9	
Chromium	105	mg/kg	2.0	1	09/07/21 19:48	09/08/21 13:35	7440-47-3	
Copper	1010	mg/kg	5.1	5	09/07/21 19:48	09/08/21 13:44	7440-50-8	
Lead	22.9	mg/kg	5.1	5	09/07/21 19:48	09/08/21 13:44	7439-92-1	
Molybdenum	21.3	mg/kg	1.0	1	09/07/21 19:48	09/08/21 13:35	7439-98-7	N2
Nickel	20.6	mg/kg	0.51	1	09/07/21 19:48	09/08/21 13:35	7440-02-0	
Selenium	5.0	mg/kg	1.0	1	09/07/21 19:48	09/08/21 13:35	7782-49-2	
Silver	3.0	mg/kg	0.51	1	09/07/21 19:48	09/08/21 13:35	7440-22-4	
Zinc	1280	mg/kg	102	20	09/07/21 19:48	09/09/21 13:27	7440-66-6	
7471 Mercury								
Analytical Method: EPA 7471 Preparation Method: EPA 7471								
Pace Analytical Services - Indianapolis								
Mercury	<11.6	mg/kg	11.6	1	09/23/21 17:00	09/23/21 18:37	7439-97-6	
Percent Moisture								
Analytical Method: SM 2540G								
Pace Analytical Services - Indianapolis								
Percent Moisture	97.5	%	0.10	1		09/09/21 10:42		N2
2540G Total Percent Solids								
Analytical Method: SM 2540G								
Pace Analytical Services - Indianapolis								
Total Solids	2.5	%	0.10	1		09/09/21 10:42		N2
2540G Total Volatile Solids								
Analytical Method: SM 2540G								
Pace Analytical Services - Indianapolis								
Total Volatile Solids	56.2	%	0.10	1		09/08/21 17:37		N2
9045 pH Soil								
Analytical Method: EPA 9045								
Pace Analytical Services - Indianapolis								
pH at 25 Degrees C	8.3	Std. Units	0.10	1		09/09/21 13:09		H3
351.2 Total Kjeldahl Nitrogen								
Analytical Method: EPA 351.2 Preparation Method: EPA 351.2								
Pace Analytical Services - Indianapolis								
Nitrogen, Kjeldahl, Total	69400	mg/kg	1980	1	09/15/21 07:05	09/15/21 11:53	7727-37-9	N2

REPORT OF LABORATORY ANALYSIS

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ANALYTICAL RESULTS

Project: Fall Biosolids

Pace Project No.: 50296623

Sample: Digester #4 **Lab ID: 50296623001** Collected: 09/02/21 14:00 Received: 09/03/21 09:15 Matrix: Solid

Results reported on a "dry weight" basis and are adjusted for percent moisture, sample size and any dilutions.

Parameters	Results	Units	Report Limit	DF	Prepared	Analyzed	CAS No.	Qual
353.2 Nitrogen, NO2/NO3								
Analytical Method: EPA 353.2 Preparation Method: EPA 353.2								
Pace Analytical Services - Indianapolis								
Nitrogen, NO2 plus NO3	<202	mg/kg	202	1	09/13/21 08:56	09/13/21 11:50		N2
Nitrogen, Nitrate	<202	mg/kg	202	1	09/13/21 08:56	09/13/21 11:50	14797-55-8	N2
4500 Ammonia Soil, Distilled								
Analytical Method: SM 4500-NH3 G Preparation Method: SM 4500-NH3 B								
Pace Analytical Services - Indianapolis								
Nitrogen, Ammonia	37700	mg/kg	2000	10	09/13/21 10:56	09/13/21 14:06	7664-41-7	N2
4500 Chloride in Soil								
Analytical Method: SM 4500-Cl-E Preparation Method: SM 4500-Cl-E								
Pace Analytical Services - Indianapolis								
Chloride	5320	mg/kg	4050	1	09/13/21 08:56	09/16/21 13:27	16887-00-6	N2

REPORT OF LABORATORY ANALYSIS

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QUALITY CONTROL DATA

Project: Fall Biosolids

Pace Project No.: 50296623

QC Batch: 641464

Analysis Method: EPA 7471

QC Batch Method: EPA 7471

Analysis Description: 7471 Mercury

Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50296623001

METHOD BLANK: 2954574

Matrix: Solid

Associated Lab Samples: 50296623001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Mercury	mg/kg	<0.20	0.20	09/23/21 18:33	

LABORATORY CONTROL SAMPLE: 2954575

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Mercury	mg/kg	0.5	0.51	103	80-120	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 2954576 2954577

Parameter	Units	50297581007 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Mercury	mg/kg	ND	0.54	0.53	0.58	0.57	102	105	75-125	1	20	

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QUALITY CONTROL DATA

Project: Fall Biosolids
Pace Project No.: 50296623

QC Batch: 640769	Analysis Method: EPA 6010
QC Batch Method: EPA 3050	Analysis Description: 6010 MET
	Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50296623001

METHOD BLANK: 2951269 Matrix: Solid

Associated Lab Samples: 50296623001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Aluminum	mg/kg	<45.3	45.3	09/22/21 00:14	
Calcium	mg/kg	<45.3	45.3	09/22/21 00:14	
Magnesium	mg/kg	<45.3	45.3	09/22/21 00:14	
Phosphorus	mg/kg	ND	45.3	09/22/21 00:14	N2
Potassium	mg/kg	<45.3	45.3	09/22/21 00:14	
Sodium	mg/kg	<45.3	45.3	09/22/21 00:14	

LABORATORY CONTROL SAMPLE: 2951270

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Aluminum	mg/kg	248	245	99	80-120	
Calcium	mg/kg	248	264	106	80-120	
Magnesium	mg/kg	248	255	103	80-120	
Phosphorus	mg/kg	124	128	103	80-120	N2
Potassium	mg/kg	248	246	99	80-120	
Sodium	mg/kg	248	268	108	80-120	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 2951271 2951272

Parameter	Units	50297670003 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Aluminum	mg/kg	2400	247	229	3760	3550	552	500	75-125	6	20	P6
Calcium	mg/kg	104000	247	229	103000	179000	-88	32800	75-125	53	20	2d,P6
Magnesium	mg/kg	20300	247	229	15200	40600	-2070	8820	75-125	91	20	2d,P6
Phosphorus	mg/kg	195	124	115	325	294	105	86	75-125	10	20	N2
Potassium	mg/kg	467	247	229	1090	935	250	204	75-125	15	20	M3
Sodium	mg/kg	198	247	229	480	482	114	124	75-125	0	20	

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QUALITY CONTROL DATA

Project: Fall Biosolids
Pace Project No.: 50296623

QC Batch: 638815	Analysis Method: EPA 6020
QC Batch Method: EPA 3050B	Analysis Description: 6020 MET
	Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50296623001

METHOD BLANK: 2941735 Matrix: Solid

Associated Lab Samples: 50296623001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Arsenic	mg/kg	<0.10	0.10	09/08/21 11:44	
Cadmium	mg/kg	<0.050	0.050	09/08/21 11:44	
Chromium	mg/kg	<0.20	0.20	09/08/21 11:44	
Copper	mg/kg	<0.10	0.10	09/08/21 11:44	
Lead	mg/kg	<0.10	0.10	09/08/21 11:44	
Molybdenum	mg/kg	<0.10	0.10	09/08/21 11:44	N2
Nickel	mg/kg	<0.050	0.050	09/08/21 11:44	
Selenium	mg/kg	<0.10	0.10	09/08/21 11:44	
Silver	mg/kg	<0.050	0.050	09/08/21 11:44	
Zinc	mg/kg	<0.50	0.50	09/08/21 11:44	

LABORATORY CONTROL SAMPLE: 2941736

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Arsenic	mg/kg	4	3.8	94	80-120	
Cadmium	mg/kg	4	4.2	105	80-120	
Chromium	mg/kg	4	3.9	96	80-120	
Copper	mg/kg	4	3.8	96	80-120	
Lead	mg/kg	4	4.2	106	80-120	
Molybdenum	mg/kg	4	4.2	106	80-120	N2
Nickel	mg/kg	4	3.8	94	80-120	
Selenium	mg/kg	4	3.8	96	80-120	
Silver	mg/kg	4	4.3	106	80-120	
Zinc	mg/kg	4	3.6	90	80-120	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 2941737 2941738

Parameter	Units	50296367002 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Arsenic	mg/kg	9.4	4	3.9	14.4	15.5	123	156	75-125	8	20	IR, M0
Cadmium	mg/kg	<0.25	4	3.9	3.9	3.8	93	95	75-125	2	20	
Chromium	mg/kg	12.6	4	3.9	15.4	16.9	68	109	75-125	9	20	M0
Copper	mg/kg	17.5	4	3.9	17.2	22.5	-8	128	75-125	27	20	1d, P6
Lead	mg/kg	25.8	4	3.9	23.0	21.6	-69	-107	75-125	6	20	P6
Molybdenum	mg/kg	2.9	4	3.9	3.5	3.2	15	7	75-125	9	20	IR, M3, N2
Nickel	mg/kg	25.6	4	3.9	35.0	38.7	231	337	75-125	10	20	P6

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QUALITY CONTROL DATA

Project: Fall Biosolids

Pace Project No.: 50296623

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 2941737 2941738												
Parameter	Units	50296367002 Result	MS	MSD	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max	
			Spike Conc.	Spike Conc.							RPD	Qual
Selenium	mg/kg	1.2	4	3.9	3.9	3.7	66	62	75-125	6	20	IR,M3
Silver	mg/kg	<0.25	4	3.9	3.8	3.7	93	94	75-125	2	20	
Zinc	mg/kg	62.3	4	3.9	65.9	78.2	88	408	75-125	17	20	P6

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QUALITY CONTROL DATA

Project: Fall Biosolids

Pace Project No.: 50296623

QC Batch: 639200

QC Batch Method: SM 2540G

Analysis Method: SM 2540G

Analysis Description: Dry Weight/Percent Moisture

Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50296623001

SAMPLE DUPLICATE: 2943015

Parameter	Units	50296623001 Result	Dup Result	RPD	Max RPD	Qualifiers
Percent Moisture	%	97.5	97.4	0	5	N2

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QUALITY CONTROL DATA

Project: Fall Biosolids

Pace Project No.: 50296623

QC Batch: 639199

QC Batch Method: SM 2540G

Analysis Method: SM 2540G

Analysis Description: 2540G Total Solids

Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50296623001

SAMPLE DUPLICATE: 2943011

Parameter	Units	50296623001 Result	Dup Result	RPD	Max RPD	Qualifiers
Total Solids	%	2.5	2.6	5	10	N2

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QUALITY CONTROL DATA

Project: Fall Biosolids

Pace Project No.: 50296623

QC Batch: 639201

Analysis Method: SM 2540G

QC Batch Method: SM 2540G

Analysis Description: 2540G Total Volatile Solids

Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50296623001

SAMPLE DUPLICATE: 2943016

Parameter	Units	50296623001 Result	Dup Result	RPD	Max RPD	Qualifiers
Total Volatile Solids	%	56.2	55.6	1	10	N2

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QUALITY CONTROL DATA

Project: Fall Biosolids

Pace Project No.: 50296623

QC Batch: 639270

Analysis Method: EPA 9045

QC Batch Method: EPA 9045

Analysis Description: 9045 pH

Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50296623001

SAMPLE DUPLICATE: 2943250

Parameter	Units	50296343001 Result	Dup Result	RPD	Max RPD	Qualifiers
pH at 25 Degrees C	Std. Units	4.4	4.4	0	2	H3

SAMPLE DUPLICATE: 2943386

Parameter	Units	50296623001 Result	Dup Result	RPD	Max RPD	Qualifiers
pH at 25 Degrees C	Std. Units	8.3	8.4	0	2	H3

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QUALITY CONTROL DATA

Project: Fall Biosolids

Pace Project No.: 50296623

QC Batch: 639984

Analysis Method: EPA 351.2

QC Batch Method: EPA 351.2

Analysis Description: 351.2 TKN

Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50296623001

METHOD BLANK: 2946837

Matrix: Solid

Associated Lab Samples: 50296623001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Nitrogen, Kjeldahl, Total	mg/kg	<49.8	49.8	09/15/21 11:44	N2

LABORATORY CONTROL SAMPLE: 2946838

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Nitrogen, Kjeldahl, Total	mg/kg	493	523	106	90-110	N2

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 2946839 2946840

Parameter	Units	50296985001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Nitrogen, Kjeldahl, Total	mg/kg	56100	3280	3220	60700	60500	140	136	90-110	0	20	M3, N2

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QUALITY CONTROL DATA

Project: Fall Biosolids

Pace Project No.: 50296623

QC Batch: 639688

Analysis Method: EPA 353.2

QC Batch Method: EPA 353.2

Analysis Description: 353.2 Nitrate + Nitrite

Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50296623001

METHOD BLANK: 2945841

Matrix: Solid

Associated Lab Samples: 50296623001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Nitrogen, Nitrate	mg/kg	<5.0	5.0	09/13/21 11:39	N2
Nitrogen, NO2 plus NO3	mg/kg	<5.0	5.0	09/13/21 11:39	N2

LABORATORY CONTROL SAMPLE: 2945842

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Nitrogen, Nitrate	mg/kg	9.9	9.9	100	90-110	N2
Nitrogen, NO2 plus NO3	mg/kg	19.8	19.6	99	90-110	N2

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 2945843 2945844

Parameter	Units	50296651001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Nitrogen, Nitrate	mg/kg	ND	10.7	10.7	10.9	11.0	102	103	90-110	1	20	N2
Nitrogen, NO2 plus NO3	mg/kg	ND	21.3	21.3	21.4	21.5	99	100	90-110	1	20	N2

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QUALITY CONTROL DATA

Project: Fall Biosolids

Pace Project No.: 50296623

QC Batch: 639687

Analysis Method: SM 4500-NH3 G

QC Batch Method: SM 4500-NH3 B

Analysis Description: 4500 Ammonia, Distilled

Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50296623001

METHOD BLANK: 2945837

Matrix: Solid

Associated Lab Samples: 50296623001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Nitrogen, Ammonia	mg/kg	<5.0	5.0	09/13/21 13:47	N2

LABORATORY CONTROL SAMPLE: 2945838

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Nitrogen, Ammonia	mg/kg	19.9	21.7	109	90-110	N2

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 2945839 2945840

Parameter	Units	50296623001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Nitrogen, Ammonia	mg/kg	37700	802	802	38400	39100	90	176	90-110	2	20	M0, N2

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QUALITY CONTROL DATA

Project: Fall Biosolids

Pace Project No.: 50296623

QC Batch: 640295

Analysis Method: SM 4500-Cl-E

QC Batch Method: SM 4500-Cl-E

Analysis Description: 4500 Chloride

Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50296623001

METHOD BLANK: 2948278

Matrix: Solid

Associated Lab Samples: 50296623001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Chloride	mg/kg	<99.1	99.1	09/16/21 13:25	N2

LABORATORY CONTROL SAMPLE: 2948279

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Chloride	mg/kg	198	202	102	90-110	N2

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 2948280 2948281

Parameter	Units	50296751001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Chloride	mg/kg	<3100	6200	6200	6830	6780	104	104	90-110	1	20	N2

Results presented on this page are in the units indicated by the "Units" column except where an alternate unit is presented to the right of the result.

REPORT OF LABORATORY ANALYSIS

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QUALIFIERS

Project: Fall Biosolids
Pace Project No.: 50296623

DEFINITIONS

DF - Dilution Factor, if reported, represents the factor applied to the reported data due to dilution of the sample aliquot.
ND - Not Detected at or above adjusted reporting limit.
TNTC - Too Numerous To Count
J - Estimated concentration above the adjusted method detection limit and below the adjusted reporting limit.
MDL - Adjusted Method Detection Limit.
PQL - Practical Quantitation Limit.
RL - Reporting Limit - The lowest concentration value that meets project requirements for quantitative data with known precision and bias for a specific analyte in a specific matrix.
S - Surrogate
1,2-Diphenylhydrazine decomposes to and cannot be separated from Azobenzene using Method 8270. The result for each analyte is a combined concentration.
Consistent with EPA guidelines, unrounded data are displayed and have been used to calculate % recovery and RPD values.
LCS(D) - Laboratory Control Sample (Duplicate)
MS(D) - Matrix Spike (Duplicate)
DUP - Sample Duplicate
RPD - Relative Percent Difference
NC - Not Calculable.
SG - Silica Gel - Clean-Up
U - Indicates the compound was analyzed for, but not detected.
N-Nitrosodiphenylamine decomposes and cannot be separated from Diphenylamine using Method 8270. The result reported for each analyte is a combined concentration.
Reported results are not rounded until the final step prior to reporting. Therefore, calculated parameters that are typically reported as "Total" may vary slightly from the sum of the reported component parameters.
Pace Analytical is TNI accredited. Contact your Pace PM for the current list of accredited analytes.
TNI - The NELAC Institute.

ANALYTE QUALIFIERS

1d	RPD is outside control limits due to sample non-homogeneity. 9-9-21 caw
2d	RPD is outside control limits due to sample non-homogeneity. JPK 09-22-21
H3	Sample was received or analysis requested beyond the recognized method holding time.
IR	The internal standard recovery associated with this result exceeds the upper control limit. The reported result should be considered an estimated value.
M0	Matrix spike recovery and/or matrix spike duplicate recovery was outside laboratory control limits.
M3	Matrix spike recovery was outside laboratory control limits due to matrix interferences.
N2	The lab does not hold NELAC/TNI accreditation for this parameter but other accreditations/certifications may apply. A complete list of accreditations/certifications is available upon request.
P6	Matrix spike recovery was outside laboratory control limits due to a parent sample concentration notably higher than the spike level.

REPORT OF LABORATORY ANALYSIS

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QUALITY CONTROL DATA CROSS REFERENCE TABLE

Project: Fall Biosolids

Pace Project No.: 50296623

Lab ID	Sample ID	QC Batch Method	QC Batch	Analytical Method	Analytical Batch
50296623001	Digester #4	EPA 3050	640769	EPA 6010	641070
50296623001	Digester #4	EPA 3050B	638815	EPA 6020	638995
50296623001	Digester #4	EPA 7471	641464	EPA 7471	641497
50296623001	Digester #4	SM 2540G	639200		
50296623001	Digester #4	SM 2540G	639199		
50296623001	Digester #4	SM 2540G	639201		
50296623001	Digester #4	EPA 9045	639270		
50296623001	Digester #4	EPA 351.2	639984	EPA 351.2	640107
50296623001	Digester #4	EPA 353.2	639688	EPA 353.2	639715
50296623001	Digester #4	SM 4500-NH3 B	639687	SM 4500-NH3 G	639725
50296623001	Digester #4	SM 4500-CI-E	640295	SM 4500-CI-E	640339

REPORT OF LABORATORY ANALYSIS

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CHAIN-OF-CUSTODY / Analytical Request Document
The Chain-of-Custody is a LEGAL DOCUMENT. All relevant fields must be completed. For more information, please visit <https://info.paceanalytical.com>

WO#: 50296623



Section A

Required Client Information:

Company:	City of Cadillac
Address:	1121 Pielt Rd
Cadillac, MI 49601	
Email:	ctomaszewski@cadillac-mi.net
Phone:	231-775-2368
Fax:	
Requested Due Date:	

Required Project Information:

Report To:	Cindy Tomaszewski
Copy To:	
Purchase Order #:	
Project Name:	Fail Biosolids
Project #:	

Invoice Information:

Attention:	CINDY TOMASZEWSKI
Company Name:	
Address:	
Pace Quote:	
Pace Project Manager:	melanie.booms@paceanalytical.com
Pace Profile #:	7675

Regulatory Agency

APDES / EGLE

State / Location

MI

Section B

Requested Analysis Filtered (Y/N)

MATRIX	CODE	Preservatives	Y/N
Drinking Water	DW	Unpreserved	
Water	WT	H2SO4	
Waste Water	WW	HNO3	
Product	P	HCl	
Soil/Solid	SL	NaOH	
Oil	OL	Na2S2O3	
Wipe	WP	Methanol	
Air	AR	Other	
Other	OT		
Tissue	TS		

Requested Analysis Filtered (Y/N)

Metals, Nutrients

Residual Chlorine (Y/N)

Unknown / Sludge

Section C

Additional Comments

1 DIGESTOR #4

9-2-1400

1 X

9-2-1400

1 X

9-2-1400

1 X

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Section D

Requested Analysis Filtered (Y/N)

1 DIGESTOR #4

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Section E

Requested Analysis Filtered (Y/N)

1 DIGESTOR #4

9-2-1400

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Section F

Requested Analysis Filtered (Y/N)

1 DIGESTOR #4

9-2-1400

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9-2-1400

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Sample Conditions Upon Receipt Form (SCUR)

Date/Time: <u>9.3.21</u>		Evaluated by: <u>WDC</u>		WO#: 50296623 PM: MSB Due Date: 09/20/21 CLIENT: GR-Cadillac	
Client: <u>City of Cadillac</u>		Profile ID:			
Project Manager: <u>MSB</u>					
Rush TAT Requested: YES ☒ NO ☒ Due Date:					
Lab Notified of Rush or Short Holds: YES ☒ NO ☒		Non Conformance Form Required: YES ☒ NO ☒			
Samples Received Via: FedEx ☒ UPS ☒ Client ☒ Pace Courier ☒ Other: _____					
Comments:					
Custody Seals Present and Intact:		YES	NO	NA	
Received Sample Information Form(s): Drinking Waters Only		YES	NO	NA	
USDA Regulated Soils: (AL, AR, CA, FL, GA, ID, LA, MS, NM, NY, NC, OK, OR, SC, TN, TX, WA or Puerto Rico)		YES	NO	N/A	
Short Holds Present (< 72 Hours):		YES	NO		
Samples Received in Hold:		YES	NO		
Custody Signatures Present:		YES	NO		
Collector Signature Present:		YES	NO		
Packing Material Used:		YES	NO		
Samples Collected Today and On Ice:		YES	NO	N/A	
IR Gun #: 280 281		Digital Thermometer #: 282 283			
Ice Type: WET Bagged / WET Loose BLUE NONE		1. Cooler Temp Upon Receipt: <u>0.3/1.1</u> °C			
Ice Location: TOP BOTTOM MIDDLE DISPERSED		Temp should be 0-6°C (Initial/Corrected)			
Temp Blank Received:		YES	NO		
Containers Intact:		YES	NO		
Correct Containers:		YES	NO		
Sufficient Volume:		YES	NO		
Sample pH Acceptable: All containers needing preservation are found to be in compliance with EPA recommendation pH Strip Lot #: _____ Exceptions are VOA, coliform, LLHg, O&G, or any container with a septum cap or preserved with HCl		YES	NO	N/A	
Residual Chlorine Absent: Cl ₂ Strip Lot #: _____ (SVOC/Pest 625, PCB 608, Total/Amenable Cyanide)		YES	NO	N/A	
VOA Headspace Acceptable (<6mm):		YES	NO	N/A	
Trip Blank Received: HCl MeOH TSP OTHER		YES	NO		
Comments:		2. Cooler Temp Upon Receipt: _____ °C			
		3. Cooler Temp Upon Receipt: _____ °C			
		4. Cooler Temp Upon Receipt: _____ °C			